



Homework 2

Subspaces, Span & Projections

Part A — Span & Subspaces

1. **Topic Space in NLP (5 pts)**

Suppose we have two “topic vectors” in a document embedding space:

$$\mathbf{t}_1 = (1, 2, 1), \mathbf{t}_2 = (0, 1, 1).$$

- Do $\{\mathbf{t}_1, \mathbf{t}_2\}$ span all of $\mathbb{R}^3$?
No, since $\mathbb{R}^3$ is a three-dimensional space, the basis should contain 3 vectors. So, two vectors cannot span the space.
- What is the dimension of their span?
Since $\mathbf{t}_1$ and $\mathbf{t}_2$ are linearly independent, the span of the two vectors is a two-dimensional space.
- Interpret: If document embeddings live in this span, what does it mean about the diversity of topics?
AI / NLP interpretation: If all document embeddings live in the span of $\{\mathbf{t}_1, \mathbf{t}_2\}$, it means every document embedding is a linear combination of the same two topic directions.
Practically: The embedding space can represent at most two independent "topics/axes of variation."
Topic diversity is limited: documents differ only by the weights on those two topics, so nuanced or orthogonal topics cannot be represented.

2. **Subspace of Neutral Sentiment (5 pts)**

Define a subspace of $\mathbb{R}^3$:

$$W = \{(x, y, z) : x + y + z = 0\}.$$

- Show W is a subspace.
By definition, the subspace W contains all vectors whose sum of the components is equal to zero.
 1. Since the zero vector $(0, 0, 0) \in \mathbb{R}^3$ also have the sum of each component equal zero, therefore $(0, 0, 0) \in W$.

2. Let $\mathbf{v}_1, \mathbf{v}_2 \in W$, such that $\mathbf{v}_1 = (x_1, y_1, z_1)$ and $\mathbf{v}_2 = (x_2, y_2, z_2)$ and $x_1 + y_1 + z_1 = 0$, $x_2 + y_2 + z_2 = 0$. Then

$$\begin{aligned} a\mathbf{v}_1 + b\mathbf{v}_2 &= a(x_1, y_1, z_1) + b(x_2, y_2, z_2) \\ &= (ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2) \end{aligned}$$

Then the sum of each component is

$$\begin{aligned} ax_1 + bx_2 + ay_1 + by_2 + az_1 + bz_2 \\ = a(x_1 + y_1 + z_1) + b(x_2 + y_2 + z_2) = a(0) + b(0) = 0 \end{aligned}$$

Since the sum of the components of $a\mathbf{v}_1 + b\mathbf{v}_2$ is equal to zero, therefore the vector $a\mathbf{v}_1 + b\mathbf{v}_2$ lies in W . Hence W is a subspace.

- Give a basis for W .

From the relation of the components of each vector (x, y, z)

$$x + y + z = 0$$

we get two free variables and one basic variable. That is $x = -y - z$. That means

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -y - z \\ y \\ z \end{bmatrix} = \begin{bmatrix} -y \\ y \\ 0 \end{bmatrix} + \begin{bmatrix} -z \\ 0 \\ z \end{bmatrix} = y \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

That is the basis is $\left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$.

- AI interpretation: If (x, y, z) = scores for (positive, neutral, negative), explain why this subspace represents a “**balanced sentiment**” space.

If $\mathbf{v} = (p, n, q)$ with $p = q$, positive and negative strengths cancel or balance out; only neutral n and the shared magnitude matter. Thus Smodels **balanced sentiment** states where positive and negative influence are equal (so the overall polarity is neutral, but intensity and neutral component still vary).

Part B — Orthogonal Projections

3. Semantic Similarity (5 pts)

Consider word embeddings:

$$\mathbf{q} = (2, 1), \mathbf{d} = (1, 1).$$

- Compute the projection of $\mathbf{q}$ onto the line spanned by $\mathbf{d}$.

$$\text{proj}_{\mathbf{d}} \mathbf{q} = \left(\frac{\mathbf{q} \cdot \mathbf{d}}{\mathbf{d} \cdot \mathbf{d}} \right) \mathbf{d} = \left(\frac{3}{2} \right) \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3/2 \\ 3/2 \end{bmatrix}$$

- Interpret: Why does cosine similarity relate to this projection?

$$\text{Cosine similarity is } \cos \theta = \left(\frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{q}\| \|\mathbf{d}\|} \right)$$

The scalar coefficient in the projection, $\frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{d}\|}$, equals $\|\mathbf{q}\| \cos \theta$.

So:

1. The length of the projection vector is $\|proj_d \mathbf{q}\| = \frac{|\mathbf{q} \cdot \mathbf{d}|}{\|\mathbf{d}\|} = \|\mathbf{q}\| \cos \theta$.
 2. Cosine similarity measures the directional alignment (angle) between $\mathbf{p}$ and $\mathbf{q}$; the projection captures how much of $\mathbf{q}$ lies in the direction of $\mathbf{d}$.
- If q is a query and d a document, what does projection tell us about relevance?
The projection tells how much of the query's meaning aligns with the document's embedding:
 1. A larger projection magnitude (or high cosine) $\rightarrow$ query content closely aligned with the document $\rightarrow$ higher relevance.
 2. The orthogonal (residual) part of $\mathbf{q}$ is the portion of the query orthogonal to the document (topics not present in the document).

4. Regression as Projection (10 pts)

Suppose we fit a linear regression model with feature matrix

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, y = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}.$$

- Compute the projection matrix $P = X(X^T X)^{-1} X^T$.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \left(\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 2/3 & -1/3 & 1/3 \\ -1/3 & 2/3 & 1/3 \\ 1/3 & 1/3 & 2/3 \end{bmatrix}$$

- Find the projected vector $\hat{\mathbf{y}} = P\mathbf{y}$.

$$\begin{bmatrix} 2/3 & -1/3 & 1/3 \\ -1/3 & 2/3 & 1/3 \\ 1/3 & 1/3 & 2/3 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 5/3 \\ 5/3 \\ 10/3 \end{bmatrix}$$

- Verify that residuals $\mathbf{y} - \hat{\mathbf{y}}$ are orthogonal to the columns of X .

$$\begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} 5/3 \\ 5/3 \\ 10/3 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 1/3 \\ -1/3 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1/3 \\ 1/3 \\ -1/3 \end{bmatrix} = \frac{1}{3} - \frac{1}{3} = 0, \quad \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1/3 \\ 1/3 \\ -1/3 \end{bmatrix} = \frac{1}{3} - \frac{1}{3} = 0$$

- AI interpretation: Explain why this “projection property” guarantees the regression line is the **best fit**.

The least-squares solution minimizes $\|\mathbf{y} - X\beta\|^2$. Geometrically, this is finding the closest point in the column space $\mathcal{C}(X)$ to $\mathbf{y}$; the perpendicular from $\mathbf{y}$ to $\mathcal{C}(X)$ gives the minimum distance. The orthogonality condition $X^T \mathbf{r} = 0$ is the

necessary and sufficient condition for minimal squared error (normal equations).
Thus $\hat{y} = Py$ is the best approximation of y within $\mathcal{C}(X)$.

Part C — AI Application Question

5. Recommender System Subspaces (5 pts)

In a recommender system, user preferences are stored as vectors in a high-dimensional space (movies $\times$ genres). Low-rank methods project users into a lower-dimensional subspace of “latent tastes.”

- Explain why using the **span of a few eigenvectors** (latent factors) gives a compact representation of users.

Why span of a few eigenvectors is compact:

1. When we compute an SVD (or eigendecomposition of a covariance/interaction matrix) and keep the top k eigenvectors (largest eigenvalues), those eigenvectors correspond to the directions of greatest variance or most important latent structure (e.g., dominant "tastes" or genre combinations).
 2. The subspace spanned by these k eigenvectors is the best rank- k linear approximation (in squared-error sense) to the original high-dimensional data. So, representing users by their coordinates in this k -dimensional latent space compresses information while preserving the most predictive structure.
 3. Practically: this reduces storage, denoises data, and captures interpretable latent factors like "action-lover," "romance-lover," etc.
- What does projecting a new user vector onto this subspace achieve?
 1. Given a new user's high-dimensional preference vector $\mathbf{u}$, projecting onto the top- k subspace yields the closest approximation $\mathbf{u}_k$ to $\mathbf{u}$ lying in the latent space. Concretely:
$$\mathbf{u}_k = U_k U_k^T \mathbf{u},$$
where U_k columns are the top k eigenvectors.
 2. This projection:
 - a) Compresses $\mathbf{u}$ to the latent coordinates $U_k^T \mathbf{u}$ (a k -vector) for storage and computation.
 - b) Removes noise and focuses on the latent tastes the recommender learned from data.
 - c) Allows efficient similarity computations and prediction (e.g., predict ratings by reconstructing in the subspace).
 - AI interpretation: Low-rank (eigenvector-based) models capture shared structure across users/items. Projecting maps each user to the common latent taste space so you can compare users or predict item affinities robustly and efficiently.